Computer Organisation and Architecture Assignment 1

1. What is Raspberry Pi?

Raspberry Pi is a small, affordable single-board computer developed to help people learn programming and electronics. Despite its compact size, it can perform many tasks such as web browsing, coding, robotics, AI projects, and IoT applications. Students and hobbyists widely use it because it is easy to set up and consumes very little power.

2. AI – Submit 10 Takeaways

1. AI enables machines to perform tasks requiring human intelligence.
2. Machine learning helps systems improve from data.
3. AI is used in healthcare, finance, education, and transport.
4. Good data improves AI accuracy.
5. AI saves time by automating repetitive work.
6. Human oversight is still important.
7. Ethical AI avoids bias and protects privacy.
8. GPUs accelerate AI training.
9. AI creates new career opportunities.
10. Continuous learning is essential in AI.

3. What is Yotta?

Yotta is an Indian company known for providing large-scale data center and cloud infrastructure services. It helps organizations securely store data, run applications, and use advanced technologies such as AI and high-performance computing.

4. What is NPTEL?

NPTEL (National Programme on Technology Enhanced Learning) is an online learning initiative by the IITs and IISc. It offers free courses in engineering, science, management, and many other subjects, with optional certification exams.

5. How do you decide which programming language to use?

The choice depends on the project goals, performance needs, platform, available libraries, developer expertise, security, maintenance, and community support. For example, Python is popular for AI, Java for enterprise software, and C++ for high-performance applications.

6. Parameters used while designing an application

Important parameters include RAM size, processor speed, storage, word size (32-bit or 64-bit), operating system, security, scalability, user interface, network requirements, and budget. These factors influence performance and user experience.

7. Why is a GPU faster than a CPU?

A CPU has a few powerful cores designed for many different tasks, while a GPU has thousands of smaller cores that perform many calculations at the same time. This parallel processing makes GPUs much faster for graphics, AI, and scientific computing.

8. Define CPU, GPU and TPU

CPU: Central Processing Unit, the main processor that executes general instructions.
GPU: Graphics Processing Unit, designed for parallel processing and graphics/AI workloads.
TPU: Tensor Processing Unit, a specialized processor created by Google to accelerate AI and machine learning tasks.

9. Last year's Turing Award

The 2025 ACM A.M. Turing Award (announced in 2026) was awarded to Andrew G. Barto and Richard S. Sutton for foundational contributions to reinforcement learning.

10. Founders of Intel and AMD

Intel was founded by Robert Noyce and Gordon Moore in 1968. AMD (Advanced Micro Devices) was founded by Jerry Sanders along with seven co-founders in 1969.

11. Which company made your laptop's CPU?

The answer depends on your laptop. Common CPU manufacturers are Intel, AMD, Apple, and Qualcomm. On a Windows laptop, you can check this by opening System Information or Task Manager → Performance → CPU.